



Shree Rahul Education Society's (Regd.)

SHREE L. R. TIWARI COLLEGE OF ENGINEERING

An Autonomous Institute Affiliated to University of Mumbai,

Approved by AICTE & DTE, Maharashtra State | NAAC Accredited, NBA Accredited Program | ISO 9001:2015 Certified |
DTE Code No: 3423 | Recognized under Section 2(f) of the UGC Act 1956 | Minority Status (Hindi Linguistic)

A Report on Mini Project

Glowify: AI-Based Personalized Skincare Recommendation System

For

Full Stack Development (Course code: 124124006)

of

Second Year (SE Sem-IV)

in

Computer Engineering

by

Team Members

Roll No.	Name
40	Ankita Rajbhar
48	Suchana Santra
56	Divya Sharma
57	Dolly Sharma

Under the guidance of

Mr. Aakash Gojare

AY 2025-2026



Shree Rahul Education Society's (Regd.)

SHREE L. R. TIWARI COLLEGE OF ENGINEERING

An Autonomous Institute Affiliated to University of Mumbai,

Approved by AICTE & DTE, Maharashtra State | NAAC Accredited, NBA Accredited Program | ISO 9001:2015 Certified |
DTE Code No: 3423 | Recognized under Section 2(f) of the UGC Act 1956 | Minority Status (Hindi Linguistic)

INDEX

Sr. No	Name Of Topic	Page No.
1	Abstract	3
2	Introduction	4
3	Problem Statement	5
4	Existing System	6
5	Proposed System	7
6	System Architecture	8
7	Methodology	9
8	Technical Requirements	10
9	Results / Output	11
10	Future Scope	12
11	Conclusion	13
12	References	14



1. Abstract

❖ Problem Overview

Skincare has become an essential aspect of personal health and lifestyle, especially among young individuals. However, selecting the right skincare routine remains a major challenge due to variations in skin type, environmental conditions, and product compositions. Most users rely on generalized advice from the internet, influencers, or trial-and-error methods, which often leads to ineffective results or even skin damage. Additionally, users lack awareness about skincare ingredients, their benefits, and safe combinations. There is also no structured way to track skincare habits or measure improvements over time. Existing systems fail to provide personalized, data-driven recommendations and do not incorporate intelligent analysis of individual skin conditions.

❖ Proposed Solution

Glowify is an AI-powered personalized skincare recommendation system designed to provide tailored skincare solutions based on individual user data. The system uses an interactive chatbot to collect user details such as skin type, concerns, and lifestyle habits. Additionally, it offers an optional face scanning module that analyzes skin conditions like acne, spots, and texture using image processing techniques.

Based on the collected data, Glowify generates customized skincare routines, including day and night regimens, ingredient recommendations, and home remedies. The platform also includes educational modules that explain skincare ingredients and their effects.

To improve user engagement and consistency, Glowify integrates habit tracking, quiz-based learning, and progress visualization features. This ensures users not only receive recommendations but also follow and monitor their skincare journey effectively.

❖ Technical Stack

Glowify is developed as a full-stack web application. The frontend is built using HTML, Tailwind CSS, and JavaScript/TypeScript to provide a responsive and interactive user interface. The backend utilizes Supabase for authentication and database management. AI functionalities include Python-based image processing for face analysis and Grok API for quiz generation. Google OAuth is integrated for secure and seamless user authentication.



Shree Rahul Education Society's (Regd.)

SHREE L. R. TIWARI COLLEGE OF ENGINEERING

An Autonomous Institute Affiliated to University of Mumbai,

Approved by AICTE & DTE, Maharashtra State | NAAC Accredited, NBA Accredited Program | ISO 9001:2015 Certified |
DTE Code No: 3423 | Recognized under Section 2(f) of the UGC Act 1956 | Minority Status (Hindi Linguistic)

2. Introduction

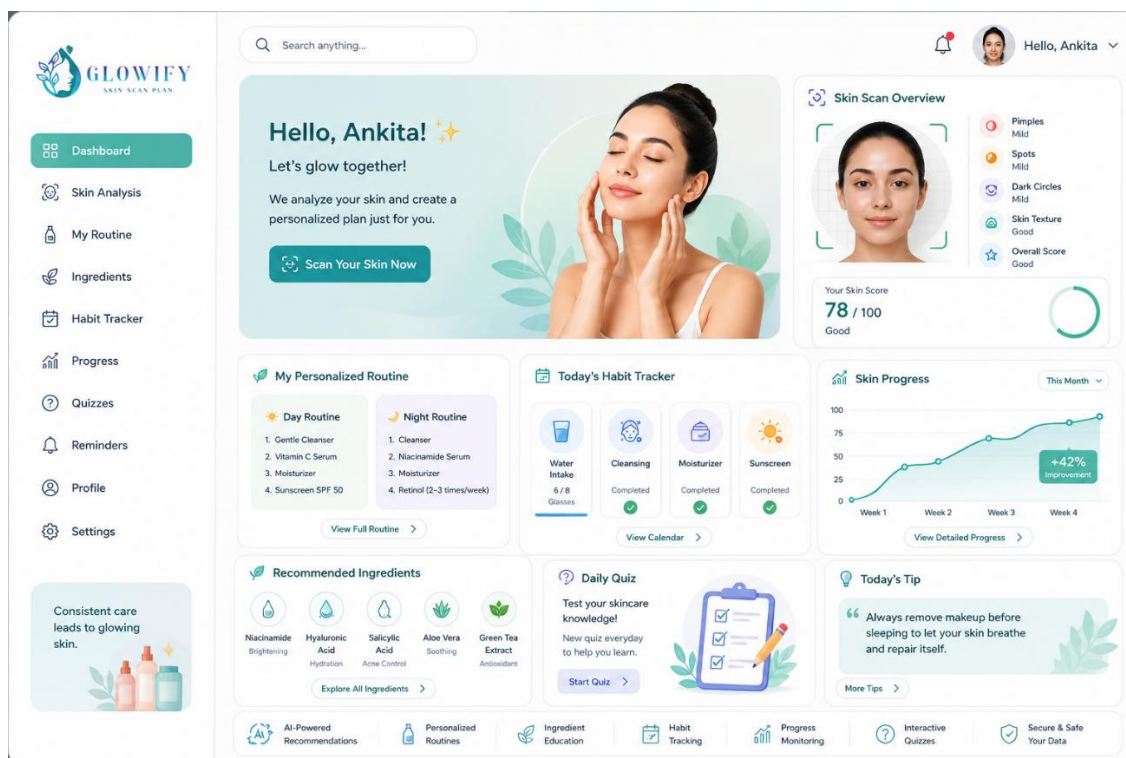
In recent years, skincare awareness has significantly increased, but many individuals still struggle to identify suitable products and routines for their specific skin conditions. The abundance of skincare products and information available online often creates confusion rather than clarity.

Traditional skincare approaches lack personalization and fail to consider individual differences such as skin type, sensitivity, and environmental exposure. Moreover, users rarely have access to tools that can analyze their skin condition or track their progress over time.

Glowify addresses these challenges by combining artificial intelligence, user interaction, and educational content into a single platform. It provides personalized skincare recommendations based on real user data rather than generic assumptions.

The system integrates multiple technologies, including chatbot-based data collection, computer vision for face analysis, and AI-driven recommendation engines. It also emphasizes user education and engagement through quizzes and progress tracking.

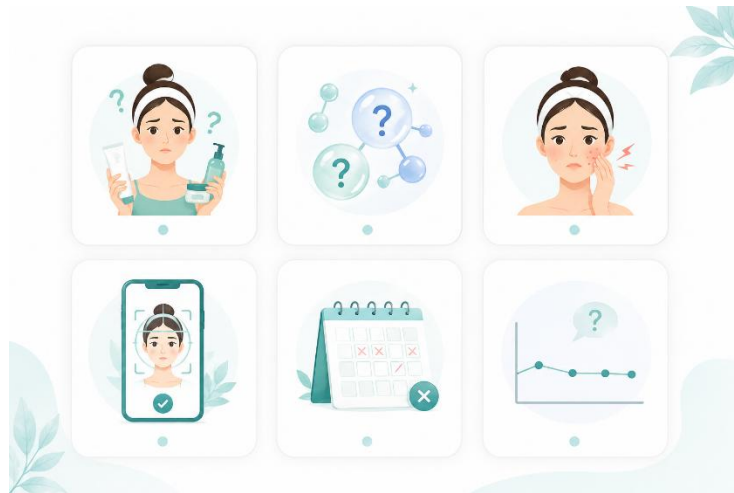
Glowify aims to make skincare more accessible, personalized, and effective by empowering users with the right knowledge and tools.





3. Problem Statement

Choosing the correct skincare routine is a complex problem due to the diversity of skin types, environmental conditions, and product formulations. Most individuals lack proper guidance and often rely on unverified sources of information.



❖ Key Problems

- 1. Lack of Personalization**
Most skincare solutions provide generic recommendations that do not consider individual skin conditions.
- 2. Limited Knowledge of Ingredients**
Users are unaware of skincare ingredients, their functions, and safe usage.
- 3. Improper Product Usage**
Incorrect use of products can lead to skin irritation or long-term damage.
- 4. Absence of Skin Analysis Tools**
Users do not have access to systems that can analyze their skin condition accurately.
- 5. No Habit Tracking Mechanism**
There is no structured way to monitor daily skincare routines.
- 6. Lack of Progress Monitoring**
Users cannot measure improvements in their skin over time.
- 7. Dependence on Trial-and-Error**
Most users experiment with multiple products without proper guidance.

These issues highlight the need for an intelligent, personalized, and user-friendly skincare system.



4. Existing System

Currently, skincare guidance is mainly obtained through blogs, YouTube tutorials, mobile applications, and dermatologist consultations. These sources provide general information about skincare routines and products, but they are not fully effective in delivering personalized solutions.

Most online platforms offer generic advice based on common skin types and do not consider individual factors such as lifestyle, environment, or specific skin concerns. Mobile apps also lack advanced analysis and often provide predefined routines. On the other hand, dermatologist consultations are accurate but expensive and not easily accessible for regular use.

Due to these limitations, users often rely on trial-and-error methods, which may lead to ineffective results or even skin damage.

❖ Limitations of Existing System

1. Generic and non-personalized recommendations
2. No integration of AI-based skin analysis
3. Lack of real-time feedback mechanisms
4. Limited user interaction and engagement
5. No habit tracking or progress monitoring
6. Expensive dermatological consultations
7. Lack of educational insights into ingredients

These limitations make existing systems inefficient for users seeking personalized and effective skincare solutions, highlighting the need for an intelligent system like Glowify.



5. Proposed System

Glowify is an AI-based skincare platform designed to provide personalized skincare solutions by combining user interaction, image analysis, and educational features. The system collects user data through an interactive chatbot, including skin type, concerns, and habits. It also offers an optional face scanning module that analyzes skin conditions such as acne, spots, and texture using image processing techniques.



Based on this data, the system generates a customized skincare routine, including day and night plans, suitable ingredients, and basic care suggestions. Glowify also focuses on educating users about skincare ingredients to promote safe and informed usage. Additionally, the platform includes features like habit tracking and progress monitoring, allowing users to maintain consistency and observe improvements over time. By integrating AI with user-friendly features, Glowify provides an efficient and personalized approach to skincare management.

❖ Key Features

- **AI-Based Personalized Recommendations**

Generates customized skincare routines based on user data.

- **Interactive Chatbot**

Collects user information in a simple and engaging way.

- **Face Scanning Module**

Analyzes skin conditions like acne and spots using image processing.

- **Ingredient Education Module**

Provides information about skincare ingredients and their benefits.

- **Quiz-Based Learning**

Enhances user knowledge through interactive quizzes.

- **Habit Tracking System**

Tracks daily skincare activities to maintain consistency.

- **Progress Monitoring Dashboard**

Displays improvements using graphs and analytics.

- **Secure Authentication**

Ensures safe login using Google OAuth.

❖ Summary

The proposed system offers a smart, personalized, and user-centric approach to skincare by combining AI technology with interactive and educational features. It not only provides effective skincare solutions but also promotes awareness, consistency, and long-term skin health.



6. System Architecture

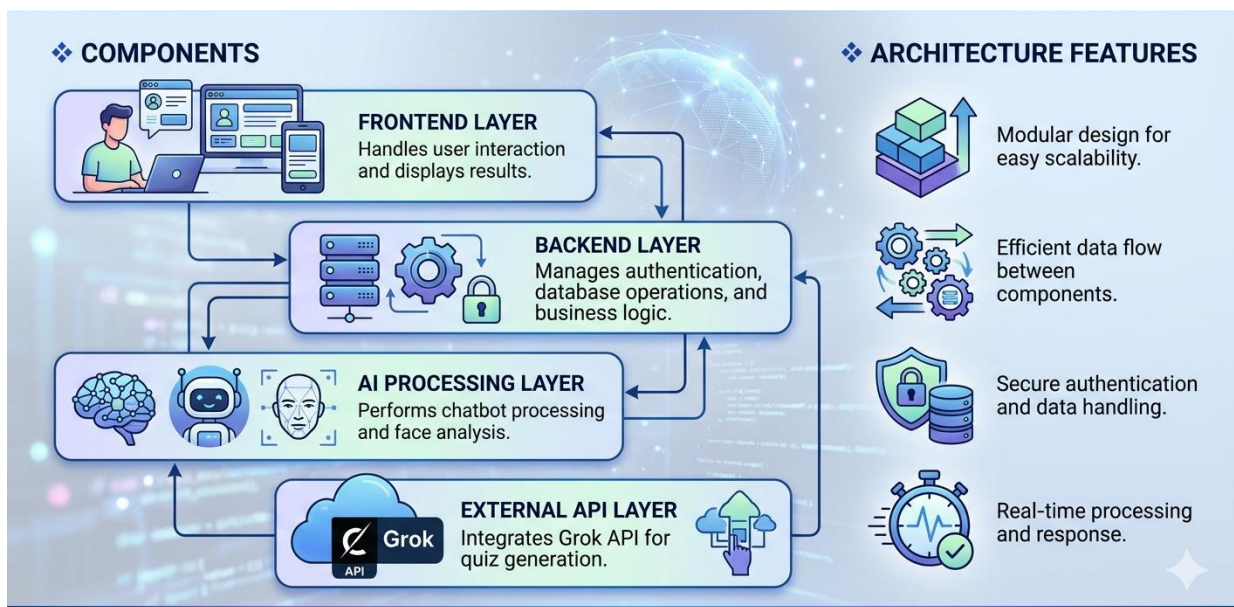
Glowify follows a modular and scalable architecture.

❖ Components

- **Frontend Layer:**
Handles user interaction and displays results.
- **Backend Layer:**
Manages authentication, database operations, and business logic.
- **AI Processing Layer:**
Performs chatbot processing and face analysis.
- **External API Layer:**
Integrates Grok API for quiz generation.

❖ Architecture Features

- Modular design for easy scalability
- Efficient data flow between components
- Secure authentication and data handling
- Real-time processing and response





7. Methodology

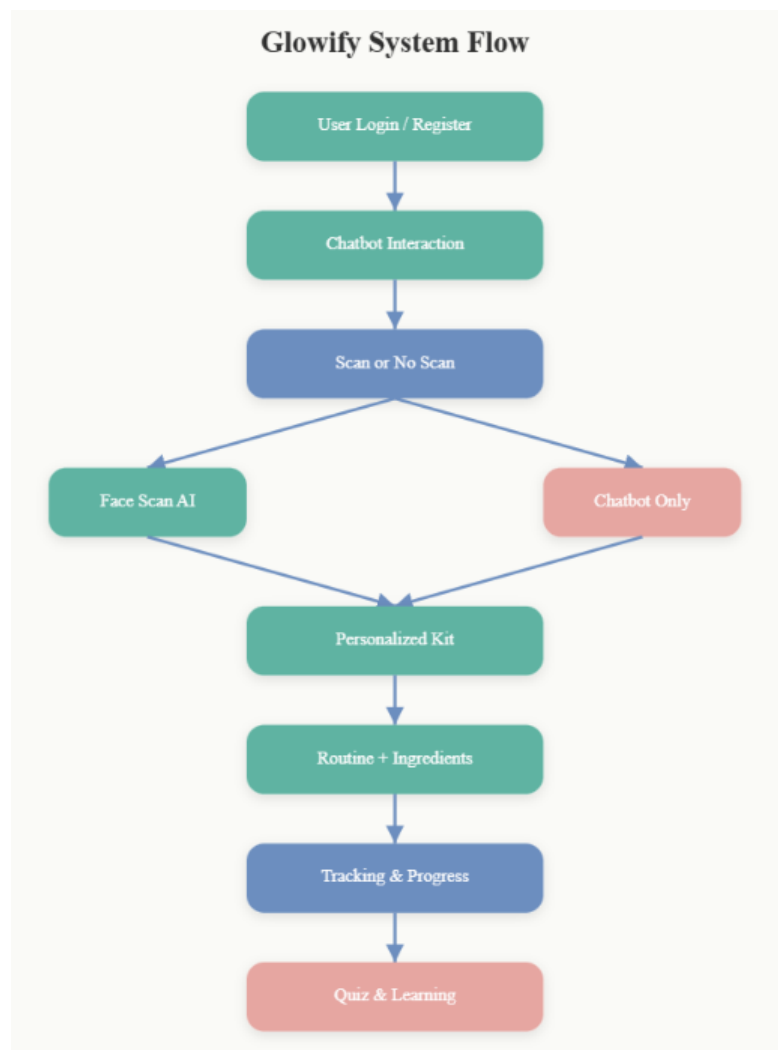
Glowify is developed using a structured methodology:

❖ Steps

1. Requirement Analysis
2. System Design
3. Frontend Development
4. Backend Development (Supabase Integration)
5. AI Module Development (Face scanning & chatbot logic)
6. Testing and Validation
7. Deployment

❖ Working Flow

- User logs in
- Chatbot collects user data
- Optional face scan is performed
- Data is processed using AI
- Personalized routine is generated
- User follows routine and logs habits
- System tracks and visualizes progress





Shree Rahul Education Society's (Regd.)

SHREE L. R. TIWARI COLLEGE OF ENGINEERING

An Autonomous Institute Affiliated to University of Mumbai,

Approved by AICTE & DTE, Maharashtra State | NAAC Accredited, NBA Accredited Program | ISO 9001:2015 Certified |
DTE Code No: 3423 | Recognized under Section 2(f) of the UGC Act 1956 | Minority Status (Hindi Linguistic)

8. Technical Requirements

❖ Software Requirements

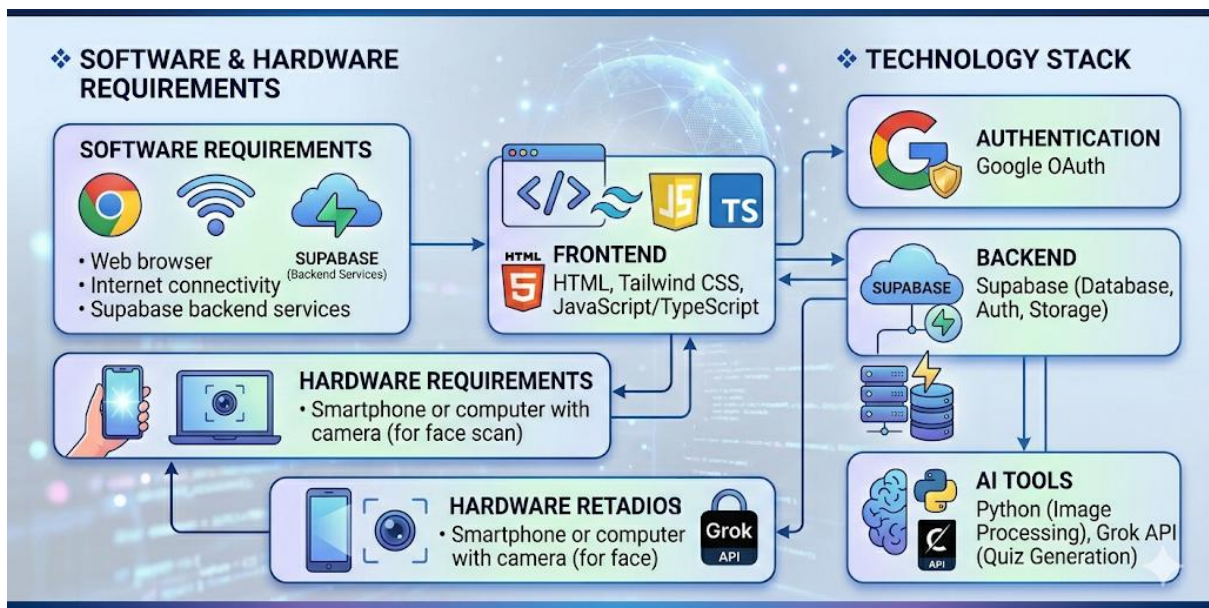
- Web browser
- Internet connectivity
- Supabase backend services

❖ Hardware Requirements

- Smartphone or computer with camera (for face scan)

❖ Technology Stack

- **Frontend:** HTML, Tailwind CSS, JavaScript/TypeScript
- **Backend:** Supabase
- **AI Tools:** Python (Image Processing), Grok API
- **Authentication:** Google OAuth





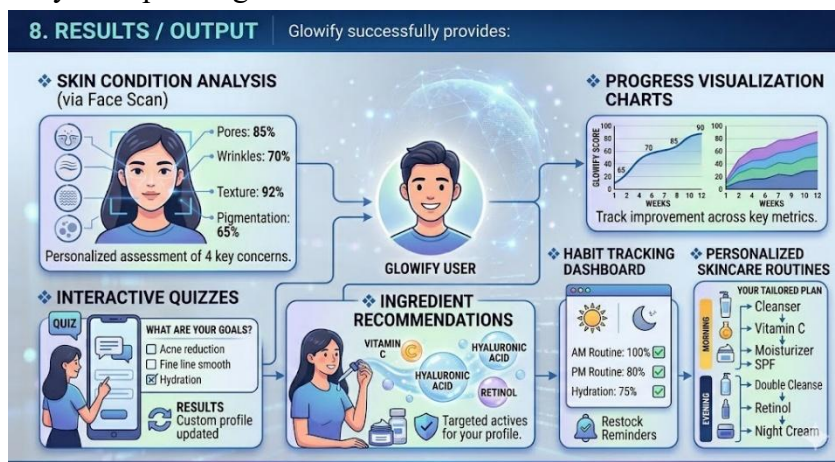
9. Results / Output

The implementation of Glowify demonstrates the successful integration of artificial intelligence, user interaction, and data-driven recommendations to provide an effective skincare solution. The system produces accurate and personalized outputs based on user input and optional facial analysis, ensuring that each user receives tailored guidance according to their specific skin type and concerns.

Glowify generates **personalized skincare routines** that include both day and night plans. These routines are customized based on factors such as skin type, common issues (acne, dryness, oiliness), and user habits. This helps users follow a structured and effective skincare regimen rather than relying on generic advice.

The **face scanning module** analyzes facial images using image processing techniques to detect visible skin conditions such as pimples, dark spots, and uneven texture. This enhances the accuracy of recommendations by incorporating real-time skin data.

The system also provides **ingredient-based recommendations**, where users receive information about suitable skincare ingredients along with their benefits and safe usage. This promotes awareness and helps users make informed decisions about the products they use.



Glowify includes **interactive quizzes** that engage users and improve their understanding of skincare concepts. These quizzes make the platform more interactive and educational.

The **habit tracking dashboard** allows users to log their daily skincare activities, such as cleansing, moisturizing, and water intake. This encourages consistency and helps users maintain a proper skincare routine. Additionally, the system offers **progress visualization charts**, which display improvements over time using graphical representations. Users can monitor their skincare journey and evaluate the effectiveness of the recommended routine.

Overall, Glowify delivers a **smooth, interactive, and user-friendly experience**. By combining personalization, analysis, and tracking, the system ensures accurate recommendations and promotes better skincare practices among users.



10. Future Scope

- **Integration with Professional Dermatologists**
Enable users to consult certified dermatologists through the platform for expert medical advice, improving reliability and trust.
- **Mobile Application Development**
Develop Android/iOS apps to increase accessibility, provide better user experience, and allow usage anytime and anywhere.
- **Advanced AI Models for Improved Accuracy**
Implement deep learning and advanced machine learning models to enhance skin analysis and provide more precise recommendations.
- **E-commerce Integration**
Allow users to directly purchase recommended skincare products within the platform, creating a complete end-to-end solution.
- **Real-Time Skin Monitoring**
Use camera-based or sensor-based technologies to continuously monitor skin condition and dynamically update recommendations.
- **Personalized Notifications and Reminders**
Provide alerts for daily skincare routines, product usage, and habit tracking to improve consistency.
- **Multi-Language Support**
Expand the platform to support multiple languages, making it accessible to a broader and more diverse audience.
- **Integration with Wearable Health Devices**
Utilize data from wearable devices (sleep, hydration, activity levels) to enhance skincare recommendations and overall analysis.



Shree Rahul Education Society's (Regd.)

SHREE L. R. TIWARI COLLEGE OF ENGINEERING

An Autonomous Institute Affiliated to University of Mumbai,

Approved by AICTE & DTE, Maharashtra State | NAAC Accredited, NBA Accredited Program | ISO 9001:2015 Certified |
DTE Code No: 3423 | Recognized under Section 2(f) of the UGC Act 1956 | Minority Status (Hindi Linguistic)

11. Conclusion

- Glowify successfully addresses the problem of generic skincare advice by providing **personalized recommendations** using AI.
- The system collects user data through chatbot interaction and optional face scanning to ensure **accurate analysis**.
- It generates customized skincare routines, including **day and night plans**, ingredient suggestions, and basic care guidance.
- The platform also focuses on **educating users** about skincare ingredients and safe practices.
- Features like **habit tracking and progress monitoring** encourage users to maintain consistency and observe improvements.
- Interactive elements such as **chatbot and quizzes** enhance user engagement and learning.
- The system offers a **user-friendly, efficient, and scalable solution** for personal skincare management.
- Overall, Glowify demonstrates the effective use of **AI and modern technologies** in improving everyday health and wellness practices.





Shree Rahul Education Society's (Regd.)

SHREE L. R. TIWARI COLLEGE OF ENGINEERING

An Autonomous Institute Affiliated to University of Mumbai,

Approved by AICTE & DTE, Maharashtra State | NAAC Accredited, NBA Accredited Program | ISO 9001:2015 Certified |
DTE Code No: 3423 | Recognized under Section 2(f) of the UGC Act 1956 | Minority Status (Hindi Linguistic)

12. References

1. **Supabase Documentation**
<https://supabase.com/docs>
2. **Python OpenCV Documentation**
<https://docs.opencv.org>
3. **OAuth 2.0 Documentation**
<https://oauth.net/2/>
4. **Grok API Documentation**
<https://console.groq.com/docs>
5. **Dermatology and Skincare Research (NIH)**
<https://www.ncbi.nlm.nih.gov/pmc/>
6. **American Academy of Dermatology (AAD)**
<https://www.aad.org/public>